

Leaf Stomata Microscope Detective

Compare stomatal density with a standardized count and make a cautious, testable hypothesis.

Win condition: Most reliable density comparison with mean, range, and an evidence-limited hypothesis.

THE SCIENCE YOU ARE USING

Stomata: pores for gas exchange. Stomata are pores that support carbon-dioxide uptake and water-vapor loss. Actual gas exchange and water use depend on aperture, pore size, conductance, species, leaf surface, and environmental conditions, not stomatal counts alone.

Sampling and counting. Use the same leaf surface, preparation, magnification, field area, and counting rule for every sample. Count at least three fields, then report the mean and range. Compare stomatal density and make a cautious hypothesis; counts alone cannot rank actual water use.

HOW TO RUN IT

- 01 **Make a safe peel.** Paint clear polish on the leaf underside, let it dry, and lift the print with clear tape. Or use a prepared slide.
- 02 **Focus the field.** Place the peel under the microscope and focus until the stomata are sharp.
- 03 **Count several fields.** Count stomata in three or more equal fields of view, then calculate the mean and range to show both center and variation.
- 04 **Compare leaves.** Repeat with the same surface, preparation, magnification, field area, and counting rule, then build a team density table.
- 05 **Compare density cautiously.** Compare mean stomatal density and range, then write a cautious hypothesis. Counts alone cannot rank actual water use; aperture, pore size, gas exchange, species, and conditions also matter.

Safety: Clear nail polish gives off fumes, so it is applied only at the ventilated instructor station, kept away from heat, and capped when not in use. Gloves are available there if you do not want polish on your fingers; no goggles are needed. Handle glass slides by the edges. Plant or latex allergy: use the prepared slides instead and latex-free gloves.

MATERIALS

Leaf samples	varied
Clear tape	shared
Clear nail polish (ventilated station) + glass slides; prepared stomata slides as backup	shared
Digital microscope (one per team)	1 per team (4; two on hand, rotate or borrow two more)
Grid sheets	1 per team (4)
Gloves	optional, shared box at the polish station

YOUR PLAN AND DATA

Use one standardized counting method and report both mean and range for every leaf.

Prediction (which sample will have the highest stomatal density, and why): _____

Leaf 1 name/type and counts per field (at least 3): _____ mean: _____ range: _____

Leaf 2 name/type and counts per field (at least 3): _____ mean: _____ range: _____

Leaf 3 name/type and counts per field (at least 3): _____ mean: _____ range: _____

Density comparison, highest to lowest mean under our fixed method: _____

DEFEND YOUR COMPARISON

What mean and range support your density comparison, and what cautious hypothesis can you make? _____

Where could counting error enter, and what do the mean and range reveal about sampling variation? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Standardized counting accuracy	30
Cautious evidence-backed hypothesis	25
Mean-and-range data table	10
Cleanup and care	10
Total	100